Second-Order Einstein–Cartan from Entanglement: The Palatini Symplectic Form and an Axial Obstruction

Robert W. McGwier
CTO, Cohere Technology Group
GitHub: n4hy

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Abstract

The remaining open of the Emergent Gravity Program that can be forced to a stop at a published standard is nonlinear Einstein–Cartan. Faulkner, Haehl, Hijano, Parrikar, Rabideau and Van Raamsdonk [2] already have Einstein's equations through second order from ball relative entropy. This note asks for the same statement with the Palatini connection independent and torsion algebraic in spin.

The Palatini symplectic potential is $\theta = (1/2\kappa)\,\varepsilon_{abcd}\, e^a \wedge e^b \wedge \delta\omega^{cd}$. It contains $\delta\omega$ and not $d(\delta e)$. Dirac coupling to ω is algebraic and adds no θ. On-shell, $\omega = \mathring{\omega}(e) + K[s]$, the Hehl–Datta term is a potential, and the reduced pre-symplectic form is that of Einstein–Hilbert plus Dirac. Second-order metric dynamics therefore collapse to FHHPRV plus a contact source T_{HD}, *if* the only CFT data are stress-tensor one- and two-point functions.

The axial channel does not match. CFT Fisher information for a conserved current is nonlocal (a modular-smeared $\langle J_5 J_5\rangle$, the same kernel that produced). Algebraic torsion has no bulk kinetic term and cannot reproduce that two-point function. This is the same mismatch Paper VI absorbed at linear order by a dictionary improvement, now visible in the symplectic form.

Verdict. At the FHHPRV standard: second-order *Einstein* gravity from entanglement stands (cited). Second-order *Einstein–Cartan*, including a generated Hehl–Datta contact as the dual of CFT axial Fisher data, is **false** without an extra propagating bulk field, which is not Einstein–Cartan. The Final Status nonlinear line remains **[O]** as a positive theorem and is closed here as an obstruction in the axial sector.

No **[O]** is moved to **[P]**.

Standard of proof

"Finish" means a statement at the grade of [2]: either the second-order EC equations (metric Einstein plus algebraic Cartan plus Hehl–Datta) are equivalent to ball relative entropy through $O(\varepsilon^2)$, or a specific matching identity fails. Jacobson's entanglement equilibrium [5] is not that grade and is not used as a substitute.

1 The FHHPRV skeleton

For CFT states prepared by a Euclidean source of strength ε, write

$$\Delta S_A - \Delta\langle H_A\rangle = -S(\rho_A\|\rho_A^{(0)}). \tag{1}$$

On a family of asymptotically AdS metrics, the Hollands–Wald identity [3] is

$$\frac{d}{d\varepsilon}\big(E_A^{\rm grav} - S_A^{\rm grav}\big) = \int_{\Sigma_A} \boldsymbol{\omega} + \int_{\Sigma_A} \mathcal{G}, \tag{2}$$

with $\boldsymbol{\omega}$ the pre-symplectic current of the gravitational Lagrangian and $\mathcal{G}$ vanishing on-shell. If S_A is represented by HRRT and E_A by the asymptotic charge, the two left-hand sides agree and

$$\frac{d}{d\varepsilon} S(\rho_A\|\rho_A^{(0)}) = \int_{\Sigma_A} \boldsymbol{\omega} + \int_{\Sigma_A} \mathcal{G}. \tag{3}$$

At first order both sides vanish except $\mathcal{G}^{(1)}$, which forces linearized Einstein [1]. At second order, for sources of stress-tensor or scalar type and $\widetilde{C}_T = a_*$, [2] compute the left-hand side from CFT two-point functions and match it to $\boldsymbol{\omega}_{\rm EH}$, forcing

$$E_{ab}^{(2)} - \tfrac{1}{2} T_{ab}^{(2)} = 0. \tag{4}$$

The only thing this note changes is the Lagrangian that defines $\boldsymbol{\omega}$ and $\mathcal{G}$.

2 Palatini data

Definition 2.1 (Palatini–NSM gravitational sector)**.**

$$S_{\rm P} = \frac{1}{4\kappa}\int \varepsilon_{abcd}\, e^a \wedge e^b \wedge R^{cd}(\omega), \qquad R^{ab} = d\omega^{ab} + \omega^a{}_c \wedge \omega^{cb}, \quad \kappa = 8\pi G. \tag{5}$$

The connection is independent. Matter couples as in the New Standard Model: fermions through ω, gauge fields not through ω.

Varying R^{ab} gives $D_\omega \delta\omega^{ab}$. After integration by parts the boundary term is

$$\theta(\delta) = \frac{1}{2\kappa}\, \varepsilon_{abcd}\, e^a \wedge e^b \wedge \delta\omega^{cd}. \tag{6}$$

There is no $d(\delta e)$ in θ. The Dirac term

$$\mathcal{L}_D[\omega] = \frac{i}{8}\, e\, \omega_\mu^{ab}\, \bar\psi\{\gamma^\mu, \gamma_{ab}\}\psi \tag{7}$$

is algebraic in ω and contributes nothing to θ.

The pre-symplectic 3-form on a slice is

$$\boldsymbol{\omega}_{\rm P}(\delta_1,\delta_2) = \frac{1}{\kappa}\, \varepsilon_{abcd}\, \delta_1 e^a \wedge e^b \wedge \delta_2\omega^{cd} - (1 \leftrightarrow 2). \tag{8}$$

Proposition 2.2 (on-shell reduction). *Write $\omega = \mathring{\omega}(e) + K$ with K the contorsion. The Cartan equation is algebraic and fixes K to the Dirac spin tensor, $K = \kappa s/2$ in the totally antisymmetric sector used by M9.1. Substituting back produces Einstein–Hilbert plus the Hehl–Datta potential*

$$\mathcal{L}_{\mathrm{HD}} = -\frac{3\kappa}{16}\, J_5^{\mu} J_{5\mu} \tag{9}$$

and no additional kinetic term for K or J_5. The reduced pre-symplectic form on (e, ψ) is

$$\boldsymbol{\omega}_{\mathrm{red}} = \boldsymbol{\omega}_{\mathrm{EH}}(\delta e, \delta e) + \boldsymbol{\omega}_{\mathrm{D}}(\delta\psi, \delta\psi). \tag{10}$$

$\mathcal{L}_{\mathrm{HD}}$ shifts the Hamiltonian. It does not enlarge the phase space.

The metric sector of (10) is exactly the $\boldsymbol{\omega}$ of [2]. Therefore:

Theorem 2.3 (metric second order, cited plus reduction). *If the only sourced CFT operator is the stress tensor (or a scalar primary) and $\widetilde{C}_T = a_*$, then (3) with $\boldsymbol{\omega}_{\mathrm{red}}$ reproduces (4) with $T^{(2)}$ including the Hehl–Datta stress of whatever axial background is already present. This is FHHPRV plus Proposition 2.2. It is not a new derivation of FHHPRV.*

3 The axial channel

Now source the CFT axial current, $\delta S_{\mathrm{CFT}} = \int \beta_\mu J_5^\mu$, as in Papers II and IV. At second order in β the relative entropy is the quantum Fisher information of J_5:

$$\delta^{(2)} S(\rho_A \| \rho_A^{(0)}) = \int_A dx \int_A dy\, \beta_\mu(x)\beta_\nu(y)\, \mathcal{F}_A^{\mu\nu}(x, y), \tag{11}$$

where $\mathcal{F}_A$ is the modular-smeared connected two-point function of J_5 — the same object whose finite-ball integral is [8]. In any unitary CFT in $d \geq 2$,

$$\langle J_5^\mu(x) J_5^\nu(y)\rangle \sim \frac{C_J}{|x-y|^{2d-2}} \tag{12}$$

is nonlocal.

On the gravity side, after Proposition 2.2, the only axial contribution to the right-hand side of (3) is the variation of the Hehl–Datta Hamiltonian. That is ultralocal:

$$\delta^{(2)} H_{\mathrm{HD}} = -\frac{3\kappa}{8} \int J_5 \cdot \delta J_5 \quad \text{with} \quad J_5 \sim \beta \text{ algebraically.} \tag{13}$$

There is no bulk Maxwell (or other propagating) symplectic current for an independent axial gauge field, because Einstein–Cartan does not have one.

Theorem 3.1 (axial obstruction). *Equations (11)–(13) cannot match for a generic smooth compactly supported β. The left-hand side of (3) is a nonlocal quadratic form (the modular Fisher kernel). The EC right-hand side is a contact. Equality for all β would require $\langle J_5(x) J_5(y)\rangle \propto \delta(x-y)$ in the ball algebra, which contradicts (12).*

This is the linear mismatch, rewritten at second order. Paper VI absorbed $\neq 0$ into a finite dictionary improvement at $O(\beta)$. That improvement is a redefinition of the *local* modular term. It does not install a propagating bulk field, and it does not make (11) ultralocal. Promoting it to a second-order equivalence is a change of theory.

Admission. Second-order Einstein–Cartan from entanglement, in the sense "ball relative entropy through $O(\varepsilon^2) \Leftrightarrow$ nonlinear EC including generated Hehl–Datta as the dual of CFT axial data," is **not a theorem**. The symplectic reduction is clean. The matching fails in the axial channel for a structural reason already visible at linear order. I did not produce a Hollands–Wald charge computation that evades Theorem 3.1. I did not derive all-orders EC. I did not repair the obstruction by adding a bulk axial gauge field: that theory is Einstein–Maxwell (or Einstein–Proca), not Einstein–Cartan.

4 What would have counted as success

A derivation in which

1. $\boldsymbol{\omega}$ is that of Palatini, not of Einstein–Maxwell;
2. $\mathcal{G} = 0$ is the second-order Einstein equation *and* the algebraic Cartan equation;
3. $\mathcal{L}_{\text{HD}}$ is generated, not inserted;
4. (11) equals $\int \boldsymbol{\omega}_{\text{red}}$ for axial β,

for all balls. Item 4 fails. Items 1–3 hold only after reducing away from the CFT current two-point function.

5 Verdict

Claim	Status
Palatini θ is (6)	derived
Algebraic Dirac–ω coupling adds no θ	derived
On-shell reduction to EH + HD potential	derived (M9.1 + Prop. 2.2)
Second-order Einstein from entanglement	cited [2]; not re-proved
Second-order EC including axial Fisher $\Leftrightarrow$ HD	**obstructed** (Thm. 3.1)
All-orders EC on general backgrounds	still **[O]**
Final Status "nonlinear regime" as a positive	unchanged, **[O]**

Forced finish: the question is answered *no* at the FHHPRV standard for the Cartan/axial sector. The metric Einstein half is someone else's theorem. That is the stop.

References

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